# 1.8 Solving Absolute Value Equations

Name _____ Period ____ Date _____

Algebra I 1.8 · Student sheet

**I can…**

- solve absolute value equations.
- solve equations with two absolute values and check for extraneous solutions.
- write absolute value equations for real-life situations.

**Vocabulary** · fill in the missing word

**Absolute value** — |a| is the _____ from a to 0 on a number line

**Absolute value equation** — an equation with a _____ inside absolute value bars

**Extraneous solution** — a value found while solving that does not satisfy the _____ equation

## Part 1 · Distance on the number line

| Question | Number(s) | Absolute value equation |
|---|---|---|
| Which numbers are 4 units from 0? | | |
| Which numbers are 3 units from 5? | | |
| Which numbers are 2 units from −1? | | |

Why do most of these have two answers? _____

## Part 2 · Notes and examples (fill in each note, then work the example)

**Absolute value equations:** |ax + b| = c means ax + b = _____ or ax + b = _____.

**No solution:** |x| = −3 has _____, because absolute value is never _____.

### Example 1

Solve each equation.

a. |x| = 6  b. |x + 3| = 10

**Isolate first:** Isolate the _____ first.

### Example 2

Solve 3|2x − 1| + 4 = 19.

**Check:** Always check for _____ solutions.

### Example 3

Solve $|x - 4| = |2x + 1|$.

### Example 4

A bag of chips is labeled 16 ounces. The actual weight can be 0.5 ounce more or less. Write and solve an equation for the least and greatest weights.

## Part 3 · Partner practice

### Partner 1

Solve $|x - 2| - 3 = 5$.

### Partner 2

How many solutions does $|x + 1| = -4$ have?